Southern New Hampshire University

CS 255 Business Requirements Document Template

System Components and Design

Purpose

- **The purpose of the DriverPass** system **is to** provide customers with a convenient way to prepare for their driving tests through online classes, practice tests, and on-the-road driving lessons. The system will also help DriverPass employees manage customers, packages, appointments, drivers, cars, and reports. The system should make it easier for DriverPass to operate its business while giving customers access to its services online.

System Background

- DriverPass is being created because many students fail their driving tests after only studying previous tests. DriverPass wants to provide better preparation through driving lessons, in-person DMV instruction, online classes, and practice tests.

- The system will be web-based and preferably hosted in the cloud. Customers will be able to create and manage their accounts and make, cancel, or modify driving lesson appointments. DriverPass employees will be able to manage customer information, appointments, drivers, cars, packages, and system accounts.
- The system will also track changes made to records so DriverPass can determine who created, changed, or canceled an appointment. The system should allow reports and other information to be downloaded for use in programs such as Excel.

Objectives and Goals

- The DriverPass system should allow customers to create accounts, select packages, and manage their personal information. Customers should be able to schedule, cancel, and modify driving lesson appointments online. The secretary should also be able to schedule appointments for customers who call or visit the office.
- The system should keep track of customers, drivers, cars, packages, and appointments. It should identify which driver and car are assigned to each lesson and help prevent scheduling conflicts. Customers should be able to access online classes and practice tests when included in their package. The system should record test scores, time taken, and test status.
- The system should provide different access rights based on employee roles. The IT officer should be able to manage accounts, reset passwords, and block access when necessary. The system should also track appointment activity and allow authorized employees to download reports and information. DriverPass should be able to disable a package when it is no longer available. The system should be accessible through computers and mobile devices.

Requirements

Nonfunctional Requirements

Performance Requirements

- The DriverPass system should be web-based and accessible through computers and mobile devices. The system should respond quickly when users log in, view information, schedule appointments, or take practice tests. It should support multiple users without significant delays.
- Customer and appointment information should be saved and updated promptly when the user is online. The system should also be available with minimal technical problems because DriverPass does not want to spend time managing technical issues.

Platform Constraints

- The system should be hosted online, preferably using a cloud-based platform. It should be accessible through common web browsers on computers and mobile devices. The system will need a database to store customer, employee, appointment, package, driver, car, and test information.
- Users should not need to install special software to access the system. Reports and other information should be downloadable in a format that can be used with programs such as Excel.

Accuracy and Precision

- The system should use individual user accounts to distinguish between customers and employees. Customer information, appointments, packages, drivers, cars, and test results should be stored accurately. The system should record the correct driver and car for each driving lesson and prevent a driver or car from being assigned to conflicting appointments.
- Test scores, time taken, and test status should also be recorded accurately. The system should record who creates, changes, or cancels an appointment. Authorized employees should be notified when incorrect or conflicting information is entered.

Adaptability

- The system should allow DriverPass to add, modify, or deactivate user accounts without changing the system code. Authorized users should be able to disable packages that are no longer available. The system should also be designed so that future features can be added without completely redesigning the system.

- Major changes to packages or system modules may require assistance from a developer or system analyst. The IT officer should have the access needed to manage user accounts, reset passwords, and block access for employees who leave DriverPass. The cloud-based system should also be able to receive updates without requiring major changes from DriverPass employees.

Security

- Users should be required to log in with their account information and password. The system should use a secure connection to protect information sent between the user's device and the system. Access to information should be restricted based on the user's role.
- The IT officer should have full access to manage employee accounts, reset passwords, and block or deactivate accounts when necessary. The system should provide an automatic password reset process for users who forget their passwords. Sensitive customer information, including payment information, should be protected. Repeated unsuccessful login attempts should be limited or temporarily blocked to help protect against brute-force attacks. The system should also keep an activity record of important account and appointment changes.

Functional Requirements

- The system shall allow customers to create accounts and select a DriverPass package.
- The system shall allow customers and the secretary to schedule, cancel, and modify driving lesson appointments.
- The system shall track customers, drivers, cars, packages, and appointments.
- The system shall provide online classes and practice tests based on the customer's package.
- The system shall record test scores, time taken, and test status.
- The system shall provide different access rights for customers and employees.
- The system shall allow the IT officer to manage accounts and reset passwords.
- The system shall track appointment changes and generate reports for authorized users.

User Interface

- The DriverPass system should have a simple web-based interface that can be accessed from computers and mobile devices. Customers should have an interface for creating an account, entering personal information, selecting packages, and managing appointments. Customers should be able to view their upcoming driving lessons and make, cancel, or modify appointments.

- Customers should also be able to access online classes and practice tests when they are included in their package. They should be able to view their test progress, including the test name, time taken, score, and status. Customers should also be able to view driver comments and lesson information.
- The secretary should have an interface for entering customer information and scheduling or modifying appointments. The secretary should be able to see which customer, driver, car, date, and time are connected to each lesson. The owner and other authorized employees should be able to view business information, customer information, appointments, packages, and reports.
- The IT officer should have an interface for managing employee accounts, resetting passwords, and blocking access. The system should also include a contact page or another way for users to contact DriverPass. Forms and messages should be clear so users understand what information they need to enter.

Assumptions

- It is assumed that users will have access to a computer or mobile device and an internet connection when they need to create or modify information. It is also assumed that DriverPass will use a cloud-based system for storing and managing its information.
- The system is being designed around DriverPass having 10 cars, with each car assigned to a driver. Each driving lesson is assumed to last two hours. Customers are expected to provide accurate personal and payment information when registering.
- It is assumed that DriverPass employees will receive enough training to use the system. The DMV is expected to provide current rules, policies, and sample test questions for DriverPass to use. It is also assumed that DriverPass will provide the correct employee roles and permissions for the system.
- The initial system is assumed to support the three packages described in the interview. Major changes or new system modules may require assistance from a developer or system analyst.

Limitations

- Users will need an internet connection to create, modify, or update information in the system. Information cannot be updated while offline because this could create duplicate or conflicting data.

- The initial system will not allow employees to freely add or remove package modules without developer assistance. The system will focus on the features and requirements identified during the interview, so future features may require additional development.
- The system will also depend on the DMV for current rules, policies, and practice test information. DriverPass will depend on its cloud provider for system availability and infrastructure. The project must also work within the available schedule, budget, and technology.

Gantt Chart

[Insert chart]

